

Varun Goud Karupothula

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EDUCATION

Vellore Institute of Technology, Amaravathi B.Tech (CSE) VIT-AP University CGPA: 8.25 Full-time	2027
Sri Chaitanya Junior College, Hyderabad 12th (Intermediate) Board of Intermediate Education, Telangana 95.1% Full-time	2022
Dr. K.K.R's Gowtham School, Hyderabad 10th (SSC) Board of Secondary Education, Telangana CGPA: 10 Full-time	2020

SKILLS

Programming: Java, Python, JavaScript

Problem Solving: 70+ DSA problems solved on LeetCode

Core CS: Data Structures & Algorithms, OOP, DBMS, Operating Systems, Computer Networks

AI & ML: TensorFlow, PyTorch, Scikit-learn, Pandas, NumPy, LLM APIs, RAG, FAISS

Web & Backend: React, HTML, CSS, FastAPI, REST APIs

Databases: PostgreSQL, MySQL, MongoDB, Supabase

Cloud & Deployment: AWS, Vercel, Render

Tools: Git, GitHub, VS Code, Vite

PROJECTS

Studyys – AI-Powered Study Assistant

[Github](#) [Live](#)

- Developed a **document-aware study assistant** with custom exam compilers, active recall simulators, and a peer-to-peer library.
- Integrated **Google Gemini** for a contextual chat engine retrieving textbook definitions.
- Implemented client-side extraction using **PDF.js** and **Web Workers**, parsing documents in **under 5 seconds** with no server load.
- Designed a **PostgreSQL** schema on **Supabase** with 15 normalized tables, custom triggers, and **Row Level Security (RLS)**.

Semantic Cache Search Engine – Cluster-Aware Document Retrieval

[Github](#) [Live](#)

- Built a **cluster-aware semantic search engine** with GMM-based cache routing, a FAISS vector index, and a React dashboard.
- Encoded queries into **384-dim embeddings** via **SentenceTransformers**, clustering with **GMM** to achieve a **35–45% cache hit rate**.
- Built a CPU-optimized **FAISS (IndexFlatIP)** index over **20,000+ documents** for sub-millisecond cosine-similarity search.
- Deployed via **Docker** and **React/Vite**; cache hits above **0.85 similarity** cut avg. API response time by **80%**.

PROVENANCE – Autonomous Multi-Hop Academic Research Agent

[Github](#) [Live](#)

- Developed a **7-node LangGraph agent** executing **1–4 multi-hop loops** to autonomously retrieve and verify academic literature.
- Integrated **OpenAlex API** to index **250M+ papers** into **768-dim FAISS** vectors using 120-word chunk windows.
- Built a **two-stage retriever** filtering top-20 vector candidates down to top-5 passages using a **PyTorch Cross-Encoder**.
- Implemented **Google Gemini NLI** to audit sentence-level citations and compute a **0–100% factual risk score** on **FastAPI**.

RESEARCH WORK

Heartbeat Arrhythmia Classification Using Hybrid 1D-CNN and LSTM with Vision-Augmented Inference 🔄

2026

- Designed a **hybrid 1D-CNN + LSTM** model for five-class (AAMI EC57) heartbeat classification on the **MIT-BIH Arrhythmia Database**, achieving **98.7% accuracy** and **97.4% macro F1-score**.
- Addressed severe class imbalance ($\approx 82\%$ Normal beats) with **inverse-frequency class weighting** instead of synthetic oversampling, using **Butterworth bandpass filtering** and R-peak-centred segmentation.
- Built a **vision-augmented inference engine** with **OpenCV** that digitises scanned or photographed **ECG chart images** into 300-point signals for real-time diagnosis.

CERTIFICATIONS

Oracle Cloud Infrastructure 2025 Certified Generative AI Professional 🔄

2025

Oracle Cloud Infrastructure 2025 Certified AI Foundations Associate 🔄

2025

Microsoft Certified: Fabric Data Engineer Associate (DP-700) 🔄

2026